

HOW FINANCIAL PLANNING IMPACTS STARTUP SURVIVAL

A Case Study & Numerical Classroom Activity for B.Tech Fifth Semester (MDM) Students

Objective	To examine how disciplined financial planning (or the lack of it) affects a startup's ability to survive its early growth stages.
Format	Case narrative + guided numerical worksheet across 11 startup-finance metrics.
Time Needed	20–25 minutes

PART A — THE CASE

Read time: 3–4 minutes

"Two Founders, Two Paths"

Aarav and Rhea were classmates who graduated with the same idea: build a low-cost computer-vision system to help small factories catch product defects automatically. They split ₹12 lakhs between them and started two separate companies — same product, same market, ₹6 lakhs each.

Aarav (InspectAI) believed in moving fast. He rented a fancy office (₹35,000/month), hired two people immediately, bought premium laptops, and kept no real budget — just a shoebox of receipts. "We'll sort out the money once the product works," he said.

Rhea (QCEdge) moved slower but smarter. She worked from a cheap shared space (₹8,000/month), hired just one person, and tracked every rupee on a simple weekly spreadsheet — costs, cash left, and how many months she could survive.

By month 4, Aarav had built a more polished prototype. But when someone asked how much runway he had left, he guessed "three or four months." The real number was six weeks.

Rhea's spreadsheet said four months, and she was right.

Then both companies landed the same opportunity: a factory offered a 6-week unpaid trial before signing a paid contract. A shipment delay hit both companies in week 3.

- Aarav had no buffer. He missed payroll. His team left. InspectAI shut down in month 7.
- Rhea had already planned for delays like this. She stayed calm, kept her one employee paid, and used the extra time to fix a bug. Her trial converted into a paying client in month 8 — the same client Aarav had lost.

PART B — THE NUMBERS BEHIND THE STORY

Before you calculate anything, study the financial snapshots below. Every metric in Part C is built from this data — so read it carefully.

B1. Cash & Burn Snapshot — as it stood in Month 4

(This is the exact moment Aarav guessed his runway wrong.)

Line Item	InspectAI (₹)	Notes
Cash remaining	1,95,000	All that was left of the ₹6L
Office rent / month	35,000	Fancy private office
Salaries / month (2 staff)	80,000	Hired both on day one
Misc / tools / month	15,000	Premium software, subscriptions

Line Item	QCEdge (₹)	Notes
Cash remaining	1,60,000	All that was left of the ₹6L
Office rent / month	8,000	Shared co-working desk
Salary / month (1 staff)	25,000	Hired only when needed
Misc / tools / month	7,000	Bare essentials only

B2. Balance Sheet Snapshot — Month 4

Line Item	InspectAI (₹)	Notes
Current Assets (cash only)	1,95,000	No receivables — pre-revenue
Current Liabilities	1,30,000	Overdue payroll ₹80,000 + rent ₹35,000 + vendor dues ₹15,000
Outstanding Debt	1,50,000	Vendor credit used to buy premium laptops
Owner's Equity	6,00,000	Original founder capital invested

Line Item	QCEdge (₹)	Notes
Current Assets	1,80,000	Cash ₹1,60,000 + minor other assets ₹20,000
Current Liabilities	45,000	One month of routine payables

Line Item	QCEdge (₹)	Notes
Outstanding Debt	50,000	Small equipment loan
Owner's Equity	6,00,000	Original founder capital invested

B3. Unit Economics — After the Factory Trial (Month 8 onward)

Both companies had priced the SAME contract with the factory client at ₹50,000/month. The difference lies entirely in cost structure. (InspectAI's figures below are modelled 'as if' it had survived to sign this client — useful for a fair side-by-side comparison.)

Line Item	Shared Assumptions (₹)	Notes
Monthly contract value (MRR per client)	50,000	Same product, same price, same market
Assumed long-run monthly churn	5%	Used for lifetime/LTV planning

Line Item	InspectAI (₹)	Notes
Cost to deliver per client / month (COGS)	22,500	Premium hardware + higher overhead
Cost to acquire one client (CAC)	1,80,000	Longer sales cycle, bigger team involved
Fixed monthly operating cost	1,30,000	Same as Section B1

Line Item	QCEdge (₹)	Notes
Cost to deliver per client / month (COGS)	15,000	Lean delivery, lower overhead
Cost to acquire one client (CAC)	1,00,000	Founder-led, low-cost sales motion
Fixed monthly operating cost	40,000	Same as Section B1

B4. QCEdge's Customer Growth — Months 8, 9 and 10

Line Item	QCEdge Customer Count	Notes
Month 8	1 client signed (the factory)	MRR = ₹50,000
Month 9	2 new clients added, none lost	New MRR added = ?
Month 10	1 of the 3 existing clients churns	Customers at start of month = 3

PART C — THE NUMERICAL CHALLENGE

Work through all 11 metrics below in order. Show your working in the blank space, then answer the interpretation question in your own words — that is where the real learning happens.

1. Burn Rate
2. Runway
3. CAC
4. LTV
5. LTV:CAC
6. Gross Margin
7. MRR
8. Churn Rate
9. Current Ratio
10. Debt-to-Equity
11. Break-even
